

Product Carbon Footprint Study Report

Product Name: blower and heat exchange unit, Twl-700

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: blower and heat exchange unit, Twl-700

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1Item blower and heat exchange unit, Twl-700

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1Item blower and heat exchange unit, Twl-700.

2. System Boundary

blower and heat exchange unit production, Twl-700, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of blower and heat exchange unit, Twl-700 produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **752.110 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

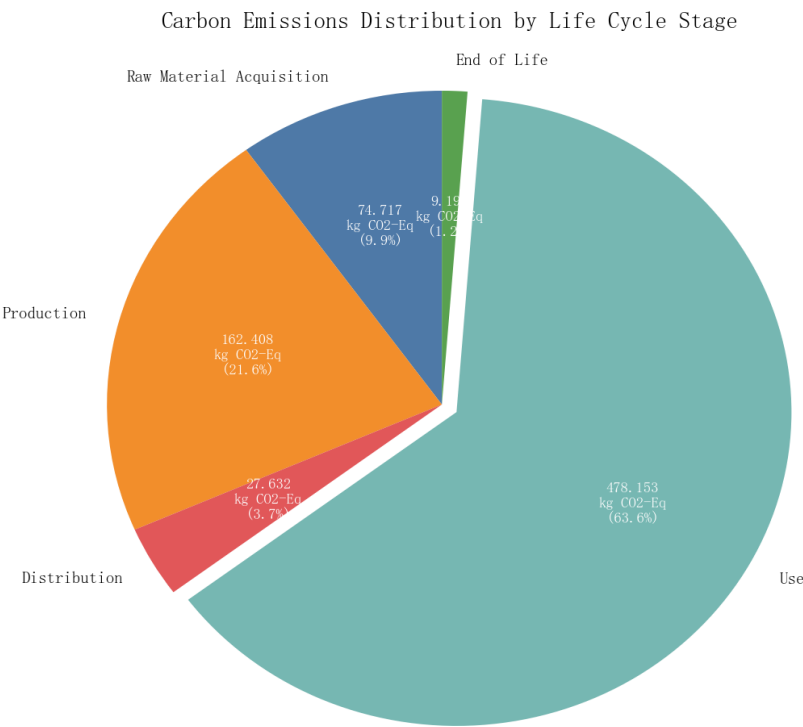


Table 1: Carbon Emissions by Life Cycle Stage

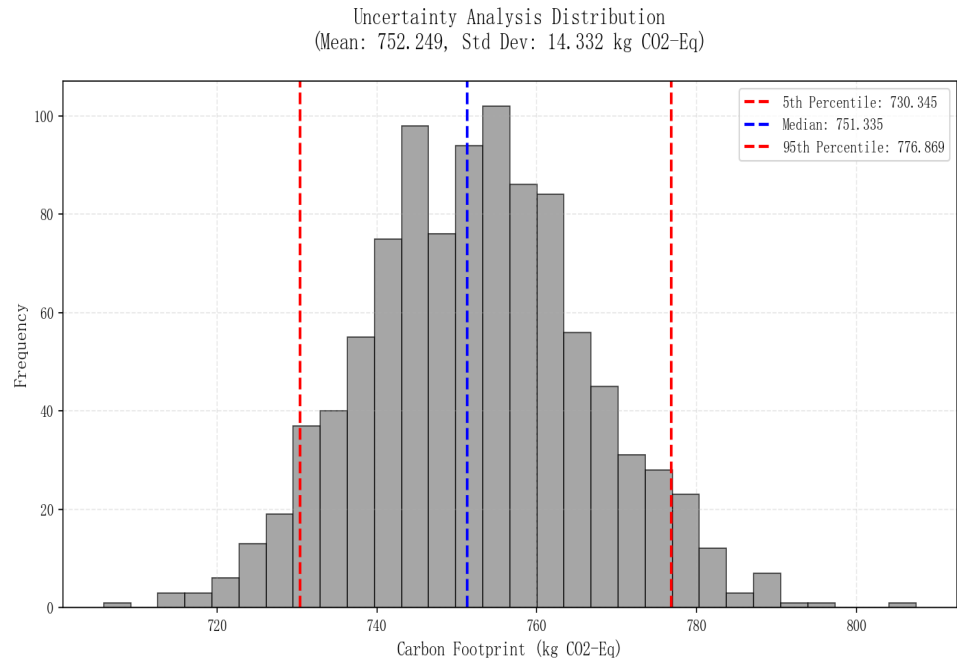
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	74.717	9.93
Production	162.408	21.59
Distribution	27.632	3.67

Use	478.153	63.57
End of Life	9.199	1.22
Total	752.110	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	752.249
Standard Deviation	14.332
Coefficient of Variation (CV)	1.91%
Median	751.335
5th Percentile	730.345
95th Percentile	776.869

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the Twl-700 blower and heat exchange unit is 752.110 kg CO₂-Eq over its full life cycle. This value reflects a moderate-to-high impact level for industrial ventilation and thermal management equipment, where operational longevity and energy intensity play decisive roles. The distribution of impacts reveals a clear dominance pattern, with the use phase accounting for 478.153 kg CO₂-Eq or 63.6% of the total, followed by production at 21.6%. Raw material extraction contributes 9.9%, while distribution and end-of-life stages remain marginal at 3.7% and 1.2%, respectively.

This contribution profile indicates that the product's environmental performance is primarily driven by in-service energy consumption rather than upstream material or manufacturing burdens. The production phase represents the second-largest source, likely stemming from energy-intensive fabrication processes such as metal forming, assembly, and component integration. Consequently, the major emission hotspots are concentrated in the use stage, where electricity demand for continuous blower and heat exchange operation dominates, and in production activities that convert raw inputs into finished assemblies.

2. Statement of Assumptions and Limitations

This assessment assumes a representative European electricity grid mix with an average emission factor, a standard 15-year service life, and typical operating hours of 4,000 hours per year under nominal load. It further presumes that end-of-life treatment follows average European waste management statistics, including recycling rates for metals and disposal for non-recyclable fractions. Transportation distances for distribution are modeled using default logistics scenarios from the producer to average European customers.

Data limitations arise from reliance on generic background datasets for raw materials and energy carriers, which may not fully capture site-specific production technologies or regional variations in supply chains. Primary data for the production stage were provided by the manufacturer but could not be independently verified for all sub-processes, introducing potential uncertainty in allocation procedures. System boundary restrictions exclude capital goods, infrastructure, and maintenance activities beyond routine component replacement, while the functional unit is defined strictly as the delivery of conditioned air over the product's defined lifetime without considering downstream system interactions.

3. Suggestions for Improvement

- Prioritize energy-efficiency redesign of the blower motor and heat exchanger to reduce use-phase electricity demand, targeting at least a 20% improvement in specific fan power.
- Implement lean manufacturing techniques and renewable electricity sourcing at the production facility to lower the 162.408 kg CO₂-Eq production burden.
- Increase recycled-content metals in raw material inputs to cut the 74.717 kg CO₂-Eq raw-material impact while maintaining structural performance.

- Optimize packaging and regional warehousing strategies to further minimize the already low distribution emissions.
- Establish primary data collection protocols for future LCAs, including sub-metered energy data and supplier-specific emission factors, to enhance overall data quality and reduce uncertainty.